Three members • Self-selected work • Course-mapped delivery • 21 October 2026

Factory Relay Three-member implementation plan

Problem establishment wrt “smart factory”: COMPLETED, as confirmed by the team. Keep that approved work as the project anchor. This plan begins with the literature survey and the implementation/evaluation work that follows.

Product Goal

Deliver and evaluate a text-first Factory Relay extension that helps two players preserve operational meaning across one reviewed language/dialect pair. Keep personal corrections and adaptation data on each user’s supported device, offer a post-session learning choice, and demonstrate play over local Wi-Fi with internet disconnected.

Build on the published standalone Factory Relay game: six scenarios, animation, guided check-back, typed messages, local HTTPS multiplayer and JSON export. New work is local translation, reviewed correction memory, eligible-device adapter training, physical-device acceptance and research evidence. The current baseline and its limits are on the next page; proposed capabilities must not be presented as completed features.

Professor’s criterion	Evidence the team will deliver
Problem establishment wrt smart factory	Completed problem sheet and approved scope; link it to every increment without reopening this stage.
Literature survey wrt problem	Traceable search log, verified comparison matrix, closest-work analysis and a defensible research gap.
Excellence and originality	Working local adaptation flow, design alternatives, precise contribution hypothesis and comparison with nearest prior work.
Effectiveness	Frozen evaluation protocol, held-out semantic/task results, resource measurements and physical offline-LAN acceptance evidence.
Expected outcomes: papers, patents, etc.	Demonstration, reproducibility package, final report/ slides and manuscript draft; conditional invention disclosure for university review.

Planning forecast: six two-week sprints, subject to the actual course deadline and team availability. No remaining teaching dates have been confirmed. Weeks start at the next agreed planning meeting. Each member initially budgets eight project hours per week, including meetings; replace these assumptions in the workbook. The roadmap also offers 8- and 15-week adaptations.

Current status · 21 October 2026

The standalone game is published. Translation, local learning and participant evaluation remain planned. Software checks do not establish communication benefits.

Dedicated project and publishing

Project root: Pure Versation/FactoryRelay. Store documents under research/ and this plan under research/team-plan. Research remains excluded from GitHub and web packages; agree an approved team-sharing location.

Resource	Recorded status and next action
GitHub · toecm/factoryrelay	Private source. Local HEAD 9b1c115 updates the README; verified game code is 6d70584. Confirm each member's access in ENG0.
Vercel · factoryrelay.vercel.app	Public game, automatically deployed from GitHub. No hosted multiplayer rooms or AI backend.
HF Space · toecm/FactoryRelay	Private Static Space, verified running. Package updates separately; no GitHub synchronization. Access is separate from GitHub.
HF dataset · toecm/factoryrelay	Public: six fictional English scenarios with answer keys. Not participant data or a dialect benchmark.

Verified capability and remaining acceptance

Available now	Remaining work / boundary
Game and export	41 automated and 114 live Vercel checks passed on 21 October; five viewports, no browser errors.
Local multiplayer	HTTPS polling. Earlier 32 checks used two browser contexts; physical devices with WAN disconnected remain unverified (ENG2).
Offline preparation	Install dependencies beforehand; keep the LAN host running on shared Wi-Fi/hotspot. Hosted pages need internet; no offline service worker.
Typed Unicode input	Translation, audio, model packs, correction memory, training and machine control are unimplemented.

Details: current-status.json and the workbook's Current Status sheet. Use the project-root setup guides. The 13 October report/manual are historical; ENG10 updates them to the delivered increment.

1 · How three members choose their work

Scrum uses one Product Owner, one Scrum Master and Developers in a self-managing team. The Product Owner orders the backlog; Developers select sprint work and decide how to deliver it. The Sprint Goal and Definition of Done guide delivery [1]. For this project, all three students contribute to implementation and research; Product Owner and Scrum Master accountabilities are additional responsibilities, not exclusive job titles.

Team-designed working agreement

- Choose one Product Owner and a different Scrum Master at kickoff. Keep one accountable person for each role during a sprint. Change accountabilities only at a sprint boundary with a clear handover; rotating demo facilitation is optional.
- Do not permanently assign one person to coding, one to literature and one to evaluation. Each member pulls an available task aligned with the Sprint Goal, records ownership, and asks another member to review it.
- Limit active implementation to one task per person and at most three team tasks in progress. Pairing is welcome. Finish or unblock work before starting more; prefer dependencies and review bottlenecks over favourite tasks.
- Balance work using capacity, complexity and learning needs. Story points are draft relative sizes, not hours, individual grades or a productivity ranking. Re-estimate as a team before selection.
- Keep statuses visible: Backlog → Ready → In progress → Review → Done; record blockers and next actions. Use a task branch and pull request in toecm/factoryrelay, with a different reviewer and evidence before merging. Treat this as a team agreement unless repository rules actually enforce it.

Event	Suggested course cadence / output
Sprint Planning	45 minutes at sprint start: agree one goal, available hours, selected stories, first tasks and reviewers.
Daily Scrum	15 minutes each working day: inspect progress toward the goal and adapt the next day's plan [1]. If the course can only meet less often, label that practice a course adaptation.
Refinement	20 minutes mid-sprint: clarify the next few stories, dependencies and estimates.
Sprint Review	30 minutes at sprint end with available supervisors: demonstrate a usable increment, inspect evidence and adjust priorities.
Retrospective	20 minutes after the review: choose one concrete improvement for the next sprint.

Use individual GitHub accounts; confirm each member's repository access in ENG0. Merge approved changes to main; the existing GitHub connection deploys Vercel automatically. Do not share owner credentials or add a separate Vercel login requirement. Invite the two teaching perspectives to review workflow realism and language/AI evidence; students select and organize the work.

2 · Six-sprint delivery forecast

Sprint	One goal and reviewable increment
S1 · Weeks 1–2	Confirm team access, language/device feasibility and physical acceptance of the existing LAN host. Demo/evidence: Show individual setup evidence, literature map and a measured local-inference request; two actual devices complete a WAN-disconnected round.
S2 · Weeks 3–4	Deliver one locally provisioned translation preview with consent and a frozen study design. Demo/evidence: Load a verified local pack, translate a synthetic message and demonstrate retain/delete choices. Present first-month review.
S3 · Weeks 5–6	Use original-plus-translated text in a real game round with analysis-ready records. Demo/evidence: Two players complete a translated round; show export versions and reviewed grouped corpus splits.
S4 · Weeks 7–8	Let a player review a session and reuse approved local corrections. Demo/evidence: Review → keep/delete → update memory → translate another example. If eligible, show a real local adapter artifact.
S5 · Weeks 9–10	Evaluate stable conditions and collect a small exploratory user pilot. Demo/evidence: Blinded condition results, pilot observations and physical device behavior; adapter promotion/rollback only if ready.
S6 · Weeks 11–12	Deliver a reproducible research extension and a defensible course evidence package. Demo/evidence: Offline end-to-end extension demo, measured tables, updated report/slides, manuscript and achieved/deferred feature matrix.

Capacity and scope control

The Team sheet calculates capacity from each member’s available hours, meeting time and a 20% uncertainty reserve. With the draft values, about 28 team hours remain for delivery per sprint. Select work using those hours and observed throughput, not a fixed points quota. Break stories into tasks of at most one member workday; move unfinished stories back for replanning without calling them Done.

The roadmap is an ordered forecast, not six fixed contracts. Every sprint includes implementation, validation and a usable demonstration; literature and writing continue alongside the increments. At the end of Sprint 1, use the device and language findings to confirm achievable scope. At the end of Sprint 2, present the first-month progress review.

Available time	Adaptation
8 weeks / 4 sprints	Keep one language pair and one translation direction if necessary. Deliver frozen-model and approved-memory conditions, physical LAN proof and a pilot report. Keep training feasibility on the backlog; defer full adapter integration and both stretch tracks if capacity fails. Record deferred requirements explicitly.
15 weeks	Keep the first six sprints, add Sprint 7 for replication or one eligible stretch feature, then one week for submission and demonstration. Do not postpone evaluation until the final week.

3 · Backlog and the first planning meeting

Use the Excel workbook as the working board. Backlog and Story Details describe future work; Tasks holds smaller checklist items. The completed baseline is recorded separately so existing game and publishing work are not estimated again. Other sheets cover sprint goals, literature, evaluation, risks and course evidence. Empty owner/reviewer fields are intentional: members choose their work.

Priority	How to use it
Core	Required for the text-first local-translation study, its evidence and course deliverables.
Conditional	Genuine adapter training and activation: attempt feasibility early, then commit implementation only when data, compatible checkpoints and hardware pass the stated gate.
Stretch	Voice/ASR or a supervised physical test rig. Pull only after core delivery is stable and relevant resources exist.
Completed baseline	Approved problem establishment, the standalone game and verified publishing are retained as evidence. Physical-device acceptance and research efficacy remain open.

First sprint: three unassigned task bundles

Open bundle — any member may choose	First concrete output
Literature and closest-work map	Search protocol, screened primary sources and an initial comparison matrix. Review another member’s feasibility findings.
Device/model feasibility	Record two available device profiles; benchmark a small licensed candidate on synthetic messages; test whether genuine local adapter training is possible on one device.
Physical LAN and measurement baseline	Use the existing game on two actual devices without internet; log reconnect/export results and propose evaluation fields.

Complete the shared ENG0 onboarding checklist alongside these bundles: confirm access, reproduce the standalone setup/checks, and agree the branch/review and evidence workflow. Choose bundles by preference and capacity, then cross-review cyclically. This example does not assign permanent specialties; pair on difficult tasks and rebalance at each sprint.

Ready-to-pull checklist

- The story has a clear user/research value, measurable acceptance criteria and a deliverable.
- Dependencies and access needs are satisfied; the scope fits the sprint and tasks are small enough to finish.
- A volunteer owner and a different reviewer have capacity; uncertainty has a timeboxed spike or an explicit gate.
- The task supports the Sprint Goal. If it does not, return it to the ordered backlog rather than starting it anyway.

4 · Implementation boundaries and gates

Planned target architecture

Target two supported laptops running the existing browser game plus a new local companion for model inference/storage. The companion does not exist yet. One laptop also hosts the existing HTTPS room server. The first training demonstration needs one eligible device. Phone inference/training is a separate capability to prove; accepting typed text in any script does not establish translation support for every dialect.

Prepare a versioned model pack, tokenizer and runtime on each supported device, verify them at startup and reuse cached files. Offline model use requires resources in advance; missing files must produce a clear setup state, never a silent cloud fallback [2]. The local companion must authenticate browser requests and restrict origins; do not expose an unrestricted inference/training service to the LAN.

Gate	Pass evidence / fallback
Language + reviewer gate · S1	One target pair/variety, each supported direction, competent reviewers and reusable data sources are documented. If coverage is weak, narrow the pair/direction and label the limit.
Runtime gate · S1–S2	Record load time, RAM/VRAM and translation timing on actual devices. Install/cold-start offline after preparation; unsupported hardware stays in message-only mode with a clear explanation.
Learning gate · S3–S4	User-approved corrections exist; source intent and target rendering are reviewed. Local memory changes retrieval; genuine LoRA changes adapter weights while the base remains frozen [3,4]. Do not label a glossary update as model training.
Adapter promotion gate · S4–S5	A real candidate adapter is saved, compared with the previous version on development validation and regression cases, and activated only between sessions after user choice. Failed/interrupted candidates leave the previous adapter intact.
Release gate · S5–S6	Physical LAN test, local-data checks, frozen evaluation, peer review and updated documentation pass. Open failures remain visible; no success claim is substituted for missing evidence.

Post-session choice to implement

Review original text and corrections → choose delete, keep, update local memory or train an eligible adapter → validate the candidate → activate or retain the previous version. Offer the choice after every session; queue examples or disable training with an explicit reason when resources or reviewed data are insufficient.

Shared game messages pass through the trusted LAN host and recipients. The proposed design keeps personal corrections, checkpoints and training logs on the owning device unless separately shared. The current game has no correction-memory or training service. Prove the companion/offline launch path before claiming restartable local AI use; preserve original wording alongside any reviewed rendering.

5 · How effectiveness will be measured

This is a future study; current evidence establishes no translation-quality or participant outcomes. Freeze the protocol before testing and keep the base model, context, decoding, hardware and check-back setting constant across adaptation conditions. The public six-scenario dataset includes answer keys: use it as a teaching/regression fixture, not an independent dialect benchmark or evaluation prompt source.

Condition	Purpose
B0 · Frozen model	Reference local model with no personal memory or adapter.
B1 · Fixed approved glossary	Simple terminology baseline; tests whether retrieval adds more than a static glossary.
B2 · Retrieved approved memory	Same model with speaker-reviewed examples; no weight updates.
B3 · Memory + personal adapter	Include only if real training and promotion gates pass; disclose a missing condition rather than simulating training.

Starter pilot budget: 120 reviewed short utterances across three high-risk families—negation, units and conditional sequence. Allocate 60 for adaptation, 20 for development/selection and 40 untouched for final testing. Group variants of a base case together; define whether the target is known-user adaptation or unseen-speaker generalization, and split speakers accordingly. Adjust counts before data collection if reviewer capacity differs; this is not a powered efficacy sample.

Measure	Definition / evidence
Primary: message critical-error rate	Messages with any wrong/omitted required critical field or unsupported critical addition ÷ evaluated messages. Define criticality in the reviewer rubric.
Diagnostic: field error rate	Wrong or omitted required fields ÷ applicable annotated fields: asset, action, quantity/unit, negation, condition and timing. Log unsupported additions separately.
Operational decision success	Correct first simulated decision ÷ decided rounds; record wrong attempts separately from simulator blocks.
Meaning and naturalness	Two competent reviewers rate blinded outputs using a written rubric; retain disagreements and adjudication. Evaluate each direction separately.
Practical performance	Median/p95 translation time with message length, peak RAM/VRAM, pack size and training duration on named devices; no universal “real-time” claim.
Offline and ownership	Cold-start prepared resources, complete a two-device session with WAN disabled, reconnect, export, update memory and reload/rollback an eligible adapter. Audit external traffic and local retention.

Suggested usability pilot: four consenting pairs (eight adults), matched task variants and counterbalanced condition order, subject to university requirements. Report descriptive results and uncertainty by pair/base case; do not treat every message as independent. Define any SAER variant explicitly. WER/CER belongs to the optional speech study, not the typed-text primary outcome. Negative or inconclusive adaptation results are still valid findings.

6 · Checklists and course deliverables

Definition of Done — applied to every selected story

- All acceptance criteria have evidence; completed subtasks alone do not make a story Done.
- A different member has reviewed the implementation or research output and reproduced the essential result.
- Applicable tests pass; original wording, ownership and offline claims match actual behavior. Defects and unsupported devices are recorded.
- Research citations have verified sources; data permissions, splits, versions and limitations are recorded. No result is invented or inferred from a screenshot.
- The increment integrates with the current game; documentation, board status and evidence links are updated. No open blocker prevents the demonstrated use case.

End-of-sprint checklist

- Demonstrate a usable increment against the Sprint Goal; show actual evidence, including failed hypotheses.
- Update the literature and nearest-work matrix when new evidence changes the design.
- Record feedback from the two teaching perspectives and adjust backlog order/capacity.
- Choose one retrospective improvement and make it visible in the next sprint.

Course submission	Minimum reviewable evidence
Literature survey	Search date/queries, inclusion rationale, verified citations, comparison of methods/datasets/limits and nearest-work gap statement.
Excellence / originality	Architecture and alternatives, explicit local-learning distinction, prototype demo and a qualified contribution claim tied to closest work.
Effectiveness	Protocol and dataset manifest, baseline comparisons, error analysis, resource/offline evidence and limitations.
Expected outcomes	Report + presentation + demonstrator + reproducibility package; a manuscript draft whose claims follow the results.
Conditional patent route	Prior-art feature comparison, dated development record and invention disclosure for university technology-transfer review. Filing or grant is not a guaranteed course deliverable.

A bounded research contribution could be evidence about when speaker-approved local adaptation preserves critical operational meaning, and when it harms it. RAG, LoRA and live translation already exist [3,4]; combining them does not establish originality. The prior-art story must test the closest overlaps before the team chooses its final claim.

7 · Risks, outputs and references

Risk	Response and scope decision
Model/dialect mismatch	Benchmark early; narrow the supported pair/direction, preserve the original and allow clarification. Record unresolved quality limits.
Training exceeds resources	Keep genuine training as a gated milestone. Preserve reviewed memory and the frozen baseline; queue data and report training as deferred, not completed.
Too little team time	Protect translation, local memory, evaluation and final evidence. Remove voice and hardware first; reforecast from actual throughput.
Data leakage / weak evaluation	Keep test cases out of training, memory and model context; hold scenario answers outside prompts; use grouped splits and independent review.
Patent/publication conflict	The game and fictional dataset are already public. Record what was disclosed and when; seek university review of any future technical contribution and its disclosure timing [5,6]. Do not infer novelty, patentability or an exhaustive search from this plan.

Expected outputs are under the team's control: a working increment, documented evidence, a reproducible study and a manuscript draft. Paper acceptance, patent filing and patent grant depend on results, review and institutional decisions. A well-explained negative result can support the report; it should not be rewritten as an improvement.

References and baseline

[1] Schwaber, K. and Sutherland, J. The Scrum Guide (November 2020). Official framework. Accessed 21 October 2026.
https://scrumguides.org/scrum-guide.html

[2] Hugging Face. Transformers installation: offline mode. Cached artifacts and local-only loading. Accessed 21 October 2026.
https://huggingface.co/docs/transformers/installation#offline-mode

[3] Hu et al. LoRA: Low-Rank Adaptation of Large Language Models (2021).
https://arxiv.org/abs/2106.09685

[4] Lewis et al. Retrieval-Augmented Generation for Knowledge-Intensive NLP Tasks (NeurIPS 2020).
https://arxiv.org/abs/2005.11401

[5] WIPO. How to Protect Inventions through Patents: conditions for patentability. Accessed 21 October 2026.
https://www.wipo.int/en/web/patents/protection

[6] WIPO. Frequently Asked Questions: Patents, confidentiality and disclosure. Accessed 21 October 2026.
https://www.wipo.int/en/web/patents/faq_patents

Evidence index: current-status.json and the workbook's Current Status sheet summarize the 21 October publication receipt, 41 automated checks, 114 live Vercel checks and earlier 32 LAN browser checks. Current local HEAD is 9b1c115; tested game code is 6d70584. The 13 October report/manual are historical evidence. Future stories require their own validation; this plan does not change the game.